

Disproof of conjecture 00000000159

Decompositions of complete graphs of prime order into distinct prime cycles

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Verdict: FALSE

1 The conjecture

Conjecture. *For every prime $p \geq 11$, the complete graph K_p can be decomposed into a union of cycles of pairwise distinct prime lengths.*

We disprove the conjecture by exhibiting a single prime p for which it fails, namely $p = 11$. The obstruction is a plain edge count and requires no structural graph theory.

2 The edge-count obstruction

Throughout, a *cycle* means a simple cycle of a simple graph, in the standard sense: a closed walk with no repeated vertices other than the starting and ending vertex. Its length is its number of edges.

Lemma 1 (Cycle lengths in K_p). *Let C be a cycle in K_p . Then the length of C equals the number of distinct vertices it uses, and*

$$3 \leq |C| \leq p.$$

Proof. A simple graph has no loops and no parallel edges, so a cycle uses at least three distinct vertices; hence its length is at least 3. A cycle in K_p uses pairwise distinct vertices of the p -element vertex set, so it uses at most p vertices. Each edge of the cycle joins two consecutive vertices, and since the vertices are distinct this is a bijection between the edges of C and its

vertices. Therefore the length of C equals its number of distinct vertices, which lies in $[3, p]$. $\square$

Corollary 2 (Available lengths for $p = 11$). *A cycle of K_{11} whose length is a prime lies in $\{3, 5, 7, 11\}$.*

Proof. By Lemma 1 the length is a prime in $[3, 11]$. The primes in that interval are 3, 5, 7, 11. $\square$

Lemma 3 (The available total is at most 26). *If L is a set of pairwise distinct primes, each an available cycle length in K_{11} , then*

$$\sum_{q \in L} q \leq 3 + 5 + 7 + 11 = 26.$$

Proof. By Corollary 2 every element of L belongs to $\{3, 5, 7, 11\}$. Since the elements are pairwise distinct, L is a subset of this four-element set, so its sum is at most the sum of the whole set, 26. $\square$

Theorem 4. *There is no decomposition of K_{11} into cycles of pairwise distinct prime lengths.*

Proof. Let the cycles of a purported decomposition have pairwise distinct prime lengths $L = \{q_1, \dots, q_r\}$. In either natural reading of “decomposed into a union of cycles”:

- if the decomposition is edge-disjoint, then the cycle lengths sum exactly to the number of edges of K_{11} , and
- if the cycles merely cover all edges, then the cycle lengths sum to at least the number of edges of K_{11} (the union has at most $\sum_i q_i$ edges, counted with multiplicity).

Thus in both cases

$$\sum_{q \in L} q \geq |E(K_{11})| = \binom{11}{2} = \frac{11 \cdot 10}{2} = 55.$$

But Lemma 3 gives $\sum_{q \in L} q \leq 26$, a contradiction. Therefore no such decomposition exists. $\square$

Corollary 5. *Conjecture 00000000159 is false, because it asserts the claim for every prime $p \geq 11$, and it already fails at $p = 11$.*

3 The failure is robust and worsens

The same count obstructs every prime in the conjectured range. Writing $\text{sop}(p) = \sum_{3 \leq q \leq p, q \text{ prime}} q$ for the total of all available prime lengths, we need

$$\text{sop}(p) \geq \binom{p}{2} = \frac{p(p-1)}{2}$$

for a decomposition to be even conceivable. The table below records the first few primes.

p	$\binom{p}{2}$	$\text{sop}(p)$	verdict
11	55	26	impossible
13	78	39	impossible
17	136	56	impossible
19	171	75	impossible
23	253	98	impossible

The inequality fails for every prime p with $11 \leq p \leq 2000$; this is verified exhaustively by `reproduce.py`. It also fails for all larger primes, since by the prime number theorem

$$\text{sop}(p) \sim \frac{p^2}{2 \ln p} \quad \text{while} \quad \binom{p}{2} = \frac{p^2}{2} \left(1 - \frac{1}{p}\right),$$

so $\text{sop}(p)/\binom{p}{2} \sim 1/\ln p \rightarrow 0$. The gap grows without bound, so the counting obstruction can never be repaired by taking p larger.

4 The strongest objection, and its rebuttal

The only way to escape the obstruction is to change the meaning of “cycle”. If “cycle” were weakened to mean a *closed trail* — a closed walk whose edges are distinct but whose vertices may repeat — then a closed trail in K_p could have length much larger than p , and the bound $|C| \leq p$ of Lemma 1 would disappear. With that weakened reading the edge count alone would no longer obstruct the claim, and the conjecture could conceivably be true.

This escape is not available. In graph theory a cycle is a simple cycle; a closed walk with repeated vertices is a closed trail, not a cycle, and the conjecture says “cycles”. Under the standard reading Lemma 1 applies and the obstruction is decisive. We record the reading we use explicitly, and we note in passing that the prime 2 is never an available length, because every cycle of a simple graph has length at least 3; this only makes the available total smaller and the obstruction stronger.

5 Scope and formalisation

The LaTeX argument above is complete. The subdirectory `lean4/` contains a Lean 4 development, using only `Std` (no `Mathlib`), which formalises precisely the arithmetic core of the obstruction:

- `validLengths = [3,5,7,11]` and the fact that every prime in `[3,11]` is a member (a finite check over `List.range 12`);
- `validLengths.sum = 26`;
- the central counting lemma: a duplicate-free list all of whose members belong to `validLengths` has sum at most 26;
- the corollary that no such list has sum 55;
- the final theorem `conjecture_00000000159_false`, which takes such a list together with the hypothesis that its sum equals the edge count $11 \cdot 10/2$ and derives `False`.

The graph-theoretic step — that the cycle lengths of a cover of K_{11} form such a list and must total at least $\binom{11}{2}$ — is not formalised; see `lean4/README.md` for the exact boundary of the formalisation. The formalised statement is not vacuous: it genuinely asserts that no list of pairwise distinct available prime lengths can reach the required total.

6 Files and reproduction

file	purpose
<code>main.tex</code>	this disproof
<code>build/main.pdf</code>	compiled PDF (produced by <code>tectonic main.tex</code>)
<code>reproduce.py</code>	verifies the count for all primes up to 2000
<code>lean4/</code>	Lean 4 formalisation, core library only, no <code>sorry</code>

Reproduce the numerical check with pure Python 3 (standard library only):

```
python3 reproduce.py
```

It prints the table for $p = 11, 13, 17, 19, 23$, lists the four available lengths at $p = 11$ with $3 + 5 + 7 + 11 = 26 < 55$, asserts that the bound fails for every prime in $[11, 2000]$, and ends with a `PASS` summary.

Status against the submission rules

Rule 3 requires a LaTeX source, a compiled PDF, and a Lean 4 project. All three are present.

- **LaTeX source:** `main.tex`.
- **Compiled PDF:** `build/main.pdf`, produced by `tectonic main.tex`. The source needs only the standard `article/amsmath/amssymb/amsthm` packages. (It was not regenerated in the offline environment in which this submission was assembled, since the TeX resource bundle could not be fetched there.)
- **Lean 4 project:** `lean4/`, built with `lake build`, using core Lean plus `Std` only (no `Mathlib`), with no `sorry`. The axiom footprint is machine-checkable via `lake env lean Check.lean`.